RNA-SEQ PROJECT

PROJECT PROGRESS REPORT

GSE190688 | High-grade serous ovarian carcinoma versus fallopian-tube epithelium

Project status: differential-expression analysis completed; visualization outputs generated by the DESeq2 workflow. Selected technical checks remain to be documented before the analysis is treated as fully finalized.

Prepared as a written progress record. It is not a clinical validation report.

Executive summary

This project re-analysed public bulk RNA-seq data from GSE190688 to compare six high-grade serous ovarian carcinoma (HGSOC) samples with five fallopian-tube epithelium (FTE) samples. The raw sequence runs were obtained from the NCBI Sequence Read Archive (SRA), processed through a paired-end alignment and gene-counting workflow, combined into a gene-by-sample count matrix, and analysed using DESeq2.

The final count matrix contained 60,675 genes across 11 samples. After the project filter requiring at least 10 counts in at least two samples, 34,303 genes remained for modelling. DESeq2 reported 10,255 genes with Benjamini-Hochberg adjusted p-value below 0.05: 5,739 had higher estimated expression in HGSOC and 4,516 had higher estimated expression in FTE. Of the significant genes, 9,241 also had an absolute log2 fold change of at least 1.

Project measure	Reported value
Dataset	GSE190688; human bulk RNA-seq
Biological groups	6 HGSOC and 5 FTE samples
SRA runs	11 paired-end runs: SRR17199566-SRR17199576
Input count matrix	60,675 genes x 11 samples
Genes retained for DESeq2	34,303
Genes with valid adjusted p-values	31,546
Significant DEGs	10,255 at padj < 0.05
Higher in HGSOC / higher in FTE	5,739 / 4,516
Significant genes with |log2FC| >= 1	9,241

1. Project objective and research question

1.1 Objective

To process public human RNA-seq reads and identify genes whose expression differs between HGSOC samples and fallopian-tube epithelium samples, using a reproducible bulk RNA-seq workflow from raw reads through statistical testing and visualization.

1.2 Research question

Which genes show statistically significant differences in estimated expression between HGSOC and FTE in GSE190688, and what is the direction and magnitude of those differences?

1.3 Why this comparison was selected

- HGSOC is a major ovarian cancer histological subtype and has a well-described biological relationship with fallopian-tube epithelium. FTE is therefore a relevant comparison tissue for this disease context.
- A tumour-versus-FTE analysis can identify expression patterns associated with the sampled tumour group relative to the selected epithelial comparator.
- The comparison supports a gene-level discovery workflow that can subsequently guide functional interpretation and independent validation.

BIOLOGICAL INTERPRETATION BOUNDARY

FTE is fallopian-tube epithelium; it is not the same as healthy ovarian tissue. HGSOC-versus-FTE expression differences may reflect malignant biology, cell-type composition, epithelial differentiation, tissue sampling and other factors. Do not describe this as a simple “ovarian cancer versus normal ovary” comparison.

2. Dataset discovery and selection

2.1 Selection criteria used for the final analysis

Criterion	Why it matters	Evidence for GSE190688
Biological relevance	A direct HGSOC group and a fallopian-tube epithelial comparator.	GSE190688 contains 6 HGSOC and 5 FTE samples.
Species and data type	Human (Homo sapiens) bulk RNA-seq / high-throughput expression profiling.	GEO records Homo sapiens and expression profiling by high-throughput sequencing.
Biological replication	More than one biological sample per condition.	Six HGSOC and five FTE samples provide replication, although the cohort remains small.
Raw-read access	Raw FASTQ sequence data available through SRA for independent processing.	The GEO Series links to SRA; BioProject lists 11 SRA experiments.
Library/platform consistency	A common sequencing platform and library strategy across the study.	Illumina HiSeq 2000; transcriptomic cDNA libraries; paired-end layout.
Metadata and traceability	Public sample titles, sample accessions, condition labels and run links.	GEO GSM records map to SRA SRX and SRR identifiers.
Computational feasibility	A cohort that can be processed as separate runs and combined after gene counting.	Eleven runs were processed through the project workflow; SRA listing reports about 106 Gbases in total.

These criteria summarize the practical and biological reasons for using GSE190688. The project archive does not contain a numerical, weighted scorecard that formally ranked GSE190688 against every GEO dataset; therefore, this report presents the selection as a criteria-based decision rather than a quantitative ranking.

3. GEO and SRA accession details

The accession levels describe different parts of the same public study: GSE is the GEO Series (study); GSM identifies an individual GEO sample; SRX identifies the SRA experiment/library; SRR identifies the sequenced run; PRJNA is the BioProject; SRP is the SRA Study; GPL identifies the sequencing platform.

Sample label	Group	GEO (GSM)	BioSample (SAMN)	SRA exp. (SRX)	SRA run (SRR)
HGSOC1	HGSOC	GSM5729031	SAMN23894007	SRX13379937	SRR17199566
HGSOC2	HGSOC	GSM5729032	SAMN23894008	SRX13379938	SRR17199567
HGSOC3	HGSOC	GSM5729033	SAMN23894009	SRX13379939	SRR17199568
HGSOC4	HGSOC	GSM5729034	SAMN23894010	SRX13379940	SRR17199569
HGSOC5	HGSOC	GSM5729035	SAMN23894011	SRX13379941	SRR17199570
HGSOC6	HGSOC	GSM5729036	SAMN23894012	SRX13379942	SRR17199571
FT1	FTE	GSM5729037	SAMN23894013	SRX13379943	SRR17199572
FT2	FTE	GSM5729038	SAMN23894014	SRX13379944	SRR17199573
FT3	FTE	GSM5729039	SAMN23894015	SRX13379945	SRR17199574
FT4	FTE	GSM5729040	SAMN23894016	SRX13379946	SRR17199575
FT5	FTE	GSM5729041	SAMN23894017	SRX13379947	SRR17199576

All 11 sample labels and run accessions are retained in the analysis metadata. The final DESeq2 matrix contains six HGSOC samples and five FTE samples. The condition labels were mapped from GEO sample metadata rather than inferred from accession number order. [1,3-6]

4. Original study and sequencing characteristics

The dataset is associated with the publication by Dou and colleagues, “HJURP Promotes Malignant Progression and Mediates Sensitivity to Cisplatin and WEE1-inhibitor in Serous Ovarian Cancer”, International Journal of

Biological Sciences (2022). The paper reports RNA sequencing of six HGSOC and five fallopian-tube tissue samples. [3]

Feature	Reported study information
Material	Human HGSOC tissue and fallopian-tube tissue samples
RNA extraction	TRIzol reagent
Library preparation	NEBNext Ultra RNA Library Prep Kit for Illumina
Library strategy/source	RNA-Seq; transcriptomic; cDNA selection
Sequencing instrument	Illumina HiSeq 2000
Read layout	Paired-end
Original study alignment	HISAT2 v2.0.5
Original study quantification	featureCounts v1.5.0-p3
Original study differential expression	DESeq2 v1.16.1

4.1 Computational environment and software versions

Component	Recorded project environment
Operating environment	Ubuntu 24.04.5 LTS under WSL2
Raw-read QC	FastQC v0.12.1
Combined QC reporting	MultiQC v1.18
Splice-aware alignment	HISAT2 v2.2.1
BAM sorting/indexing	SAMtools v1.22.1
Gene-level quantification	featureCounts / Subread v2.0.6
Statistical analysis	R v4.3.3; DESeq2 v1.42.1

RAW DATA VERSUS PROCESSED DATA

GEO lists an FPKM spreadsheet as processed data, but this project downloaded the SRA reads and independently generated gene-level raw counts. This is important because DESeq2 expects raw non-negative integer counts as model input; FPKM values are not an appropriate substitute.

5. End-to-end pipeline overview

Stage	Pipeline step	Purpose / output
01	Dataset discovery and selection	GEO series, publication, biological groups and selection criteria
02	SRA acquisition	Download the 11 run accessions from SRP350291
03	FASTQ conversion	Convert SRA run files into paired-end FASTQ mates (R1/R2)
04	Raw-read quality control	FastQC reports for individual FASTQ files
05	Combined quality review	MultiQC aggregation of available QC reports
06	Reference preparation	GRCh38 primary assembly HISAT2 index + Ensembl release 84 GTF
07	Splice-aware alignment	HISAT2 alignment of paired-end reads
08	BAM processing	Coordinate sort and index with SAMtools

Stage	Pipeline step	Purpose / output
09	Gene-level quantification	featureCounts paired-end fragment counts
10	Count matrix + metadata	One gene-by-sample matrix and matching 11-sample metadata
11	Differential expression	DESeq2 HGSOC-versus-FTE model and adjusted p-values
12	Visualization	PCA, MA, dispersion, volcano and top-gene heatmap outputs

6. Pipeline execution: purpose, work completed and outputs

Step 1 - Dataset discovery and sample metadata

What was done: the GEO Series record and linked SRA study were used to identify the 11 samples, their sample labels, biological groups and run accessions. The final comparison was set as HGSOC (six samples) versus FTE (five samples).

Why: correct biological labels are essential for defining the DESeq2 contrast. Assigning groups based only on SRR number would risk a sample-label error.

Output: a sample map and metadata file, later saved as ~/GSE190688/metadata/sample metadata.csv.

Step 2 - SRA download and FASTQ conversion

What was done: SRA run files were obtained for SRR17199566-SRR17199576 and converted into paired-end FASTQ files. SRA Toolkit commands used in the project included prefetch for run acquisition and fasterq-dump for FASTQ conversion; the exact SRA Toolkit version is not recorded in the available project summary.

Why: FASTQ files contain the nucleotide reads and their per-base quality scores, which are needed for raw-read quality assessment and alignment.

Output: paired FASTQ mates, conventionally named SAMPLE 1.fastq.gz and SAMPLE 2.fastq.gz.

Step 3 - Raw-read quality control (FastQC)

Tool: FastQC v0.12.1. Individual raw FASTQ files were assessed before alignment. The QC report categories include per-base sequence quality, per-sequence quality, GC content, adapter content, sequence duplication and basic read statistics.

Why: QC helps identify technical artefacts or unusual read properties before alignment. It does not prove that a sample is biologically correct, but it identifies read-level issues that may affect downstream results.

Output: individual FastQC HTML reports and ZIP archives, stored under the project FastQC report directory. Reports from team members were also collected for shared review.

Step 4 - Combined quality review (MultiQC)

Tool: MultiQC v1.18 was used in the current WSL project environment to aggregate available FastQC output into a single report. The report was saved as ~/GSE190688/multiqc results/GSE190688 MultiQC Report.html.

Why: a combined report makes it easier to compare samples, spot outliers and review consistency across the sequencing runs.

Step 5 - Reference genome and annotation preparation

Reference: human GRCh38 primary assembly. The project HISAT2 index prefix was recorded as ~/GSE190688/reference/grch38/genome. Its index provenance indicated Ensembl release 84. The gene annotation used for counting was ~/GSE190688/reference/Homo sapiens.GRCh38.84.gtf.

Why: alignment and quantification must use compatible genome coordinates and annotation identifiers. Using a common reference and GTF across all samples helps ensure that a gene ID represents the same feature in every sample.

Step 6 - Splice-aware read alignment (HISAT2)

Tool: HISAT2 v2.2.1. Paired-end FASTQ mates were aligned against the GRCh38 index. The shared project settings checklist specified HISAT2 -p 4. This produces genomic alignments while accommodating RNA-seq reads that span exon-exon junctions.

Why: alignment assigns each read to a genomic location, providing the coordinates required for later gene-level quantification.

Output: per-sample alignment stream and coordinate-sorted BAM output, with an alignment log under ~/GSE190688/logs/.

Step 7 - BAM sorting and indexing (SAMtools)

Tool: SAMtools v1.22.1. The shared settings checklist specified coordinate sorting with samtools sort -@ 2 and creation of a BAM index with samtools index.

Why: BAM is a compressed binary representation of alignments. Coordinate sorting organizes records by genomic position, and the .bai index enables efficient regional access.

Output: one sorted BAM per sample, named SAMPLE Aligned.sortedByCoord.out.bam, plus the corresponding .bai index.

Step 9 - Gene-level read counting (featureCounts)

Tool: featureCounts from Subread v2.0.6. The common project counting specification was -T 4 -p --countReadPairs with the Ensembl release 84 GTF. For paired-end data, --countReadPairs indicates that fragments/read pairs are counted rather than treating each mate as an independent fragment.

Why: alignment tells where reads map; featureCounts assigns fragments to annotated genes and generates the integer gene counts needed for DESeq2.

Output: one gene-count table and one .summary file per sample. The summary records categories such as Assigned, Unassigned Unmapped, Unassigned MultiMapping, Unassigned NoFeatures and Unassigned Ambiguity.

Step 10 - Count matrix construction and metadata validation

The per-sample gene count tables were combined by gene identifier into one gene-by-sample matrix. BAM files were not merged and sample counts were not averaged. The resulting matrix was saved as ~/GSE190688/counts/GSE190688 gene count matrix.tsv. Sample metadata was saved as ~/GSE190688/metadata/sample metadata.csv.

Validation recorded for the final inputs: 60,675 unique gene rows; 11 sample columns; sample names matched the metadata; count values were non-negative integers; and duplicate gene IDs were checked. This is the appropriate input structure for bulk RNA-seq count modelling.

Project note: SRR17199572 was listed as missing in an earlier preflight checklist. The final count matrix used for DESeq2 contains all 11 samples, so the missing-sample issue was resolved before the reported DESeq2 run. No zero-filled sample column was used.

Step 11 - Differential gene expression (DESeq2)

Environment: R 4.3.3 with DESeq2 v1.42.1. DESeq2 was used because it models raw integer RNA-seq counts with a negative-binomial framework, estimates sample size factors and gene-wise dispersion, and tests the specified condition contrast while adjusting for multiple comparisons.

Model element	Project setting
Input	Raw integer gene counts; 60,675 genes x 11 samples
Design formula	~ condition
Reference level	FTE
Contrast	HGSOC relative to FTE

Model element	Project setting
Low-count filter	At least 10 reads in at least 2 samples
Statistical test	DESeq2 default Wald test for the stated contrast
Multiple-testing adjustment	Benjamini-Hochberg adjusted p-value
Significance criterion	padj < 0.05
Effect-size descriptor	|log2FoldChange| >= 1 used to count larger changes

Direction: because FTE was specified as the reference, a positive log2FoldChange means higher estimated expression in HGSOC; a negative value means higher estimated expression in FTE. A gene is counted as significant only when its adjusted p-value is below 0.05.

Step 12 - Visualization and diagnostic plots

The DESeq2 workflow generated statistical diagnostic and expression plots. Variance-stabilized values from vst(dds, blind = FALSE) were used for PCA visualization; the statistical model itself was fitted using raw integer counts.

Visualization	What it shows	Why it was included
PCA by condition	Major sources of variation among the 11 samples using transformed expression values.	Checks whether samples show group-level expression structure and highlights possible outliers.
MA plot	Log2 fold change versus mean normalized expression.	Shows direction and magnitude of changes across the expression range.
Dispersion plot	Estimated gene-wise dispersion relative to mean expression and fitted trend.	Checks the dispersion estimation used by the negative-binomial model.
Volcano plot	Log2 fold change versus -log10 adjusted p-value.	Shows effect size and statistical evidence together; right and left directions distinguish HGSOC- and FTE-higher genes.
Top-30 DEG heatmap	Relative expression (row-wise z-scores) of the top significant genes across samples.	Displays sample-level patterns for a selected set of significant genes.

Reported plot outputs include PCA by condition.pdf, a separately reported PCA labeled.pdf, MA plot.pdf and dispersion plot.pdf. The DESeq2 script also writes Volcano plot.pdf and Top30 DEG heatmap.pdf. Before final submission, confirm that these last two files exist in the final deseq2 results folder and inspect every PDF directly. PC1/PC2 variance percentages and the precise visual separation of the groups should be read from the actual PCA plot; they are not inferred from DEG counts.

7. DESeq2 results

Metric	Result	Interpretation
Genes in starting matrix	60,675	All rows supplied in the validated count matrix.
Genes after low-count filter	34,303	Genes meeting the project rule of >=10 counts in >=2 samples.
Genes with non-missing padj	31,546	Genes for which an adjusted p-value was available.
Genes with padj = NA	2,757	Not counted as significant; independent filtering or insufficient test information can lead to unavailable adjusted p-values.
Significant DEGs	10,255	padj < 0.05.
Higher in HGSOC	5,739	Positive log2FoldChange and padj < 0.05.
Higher in FTE	4,516	Negative log2FoldChange and padj < 0.05.
Significant DEGs with |log2FC| >= 1	9,241	A subset of the 10,255 significant genes, not an additional group.

7.1 How to interpret a DESeq2 result row

Column	Meaning	How to read it
gene id	Ensembl/gene identifier	Maps the row to a gene annotation or symbol.
baseMean	Mean of normalized counts across samples	Approximate expression abundance; not the fold change.
log2FoldChange	Estimated log2 expression ratio for HGSOC / FTE	Positive = higher in HGSOC; negative = higher in FTE.
lfcSE	Standard error of log2 fold change	Uncertainty of the estimated change.
stat	Test statistic	Used by the statistical test.
pvalue	Unadjusted p-value	Evidence before multiple-testing correction.
padj	Benjamini-Hochberg adjusted p-value	Use this column for the stated significance threshold.

8. Visualization results: what each figure contributes

8.1 PCA

PCA compresses information from thousands of genes into principal components so the overall similarity among samples can be examined. In this project, the DESeq2 script applied a variance-stabilizing transformation before PCA. The plot should be used to discuss whether the six HGSOC and five FTE samples form distinguishable patterns, whether any sample is distant from its group, and whether one sample appears to dominate the variation. The percentage variance explained by PC1 and PC2 must be quoted from the saved figure itself.

8.2 MA plot

The MA plot places average normalized expression on the horizontal axis and log2 fold change on the vertical axis. It helps show whether fold changes are distributed around zero and whether large changes are concentrated among low- or high-expression genes.

8.3 Dispersion plot

The dispersion plot displays gene-wise dispersion estimates and the fitted mean-dispersion trend. RNA-seq biological replicates vary in expression, and DESeq2 uses dispersion estimates in its negative-binomial model. This figure is a model diagnostic, not a ranking of important genes.

8.4 Volcano plot

The volcano plot combines log2 fold change and adjusted p-value. Genes on the positive side have higher estimated expression in HGSOC; genes on the negative side have higher estimated expression in FTE. The horizontal significance guide corresponds to padj = 0.05; vertical guides at log2 fold changes of -1 and +1 indicate two-fold effect-size boundaries where displayed.

8.5 Top-30 DEG heatmap

The heatmap shows relative expression patterns for the selected top significant genes. Row-wise z-scores emphasize whether each gene is relatively higher or lower across samples; they are not raw counts and do not show absolute expression levels. The sample dendrogram or grouping should be checked to see whether the pattern is consistent with condition and whether an individual sample drives it.

9. Quality-control observations and items requiring final audit

9.1 Documented sample-level example: SRR17199574 / FT3

The saved project checklist records an overall HISAT2 alignment rate of 93.11% for SRR17199574 and featureCounts Assigned = 27,278,776, reported as 73.3% for that run. These are different measures with different

denominators: HISAT2 alignment describes read mapping, while featureCounts assignment describes assignment to annotated features. They should not be directly compared as if they were the same statistic.

9.2 Sample requiring attention: SRR17199571 / HGSOC6

The project notes record a lower gene-count total for SRR17199571 (13,672,175) than for the other samples (approximately 20-27 million). Its featureCounts summary included substantial counts in unmapped, multimapping, no-feature and ambiguity categories. Alignment was repeated and the same general result was obtained; the sample was reported to remain with its HGSOC group in the PCA. It was retained in the 11-sample analysis, but its mapping and assignment metrics should remain a documented QC consideration.

12. Progress status at the visualization stage

Stage	Status / comment
Dataset discovery and selection	Completed for final analysis dataset GSE190688; earlier cell-line scoping documented separately.
GEO/SRA metadata mapping	Completed; 11 samples mapped to HGSOC or FTE.
SRA retrieval and FASTQ conversion	Completed for the 11-run project; exact SRA Toolkit version not retained in the project summary.
Raw FastQC / MultiQC	Reports generated; review of every sample's flags should be retained in the QC record.
Reference and annotation preparation	GRCh38 / Ensembl release 84 baseline recorded.
HISAT2 alignment and BAM processing	Completed for the final analysis cohort; complete per-sample command sign-off still needs audit.
featureCounts quantification	Per-sample count tables and summaries produced; strandedness and exact settings need final command-level confirmation.
Count matrix and metadata	Completed and validated: 60,675 genes x 11 samples.
DESeq2	Completed; 10,255 significant DEGs at padj < 0.05.
Visualization	PCA, MA, dispersion and volcano/heatmap plot generation in the DESeq2 workflow; inspect final saved files before presenting.

13. References and data sources

[1] NCBI Gene Expression Omnibus. GSE190688: RNA sequencing of 6 HGSOC and 5 FTE. GEO Series record. Open source

[2] NCBI BioProject. PRJNA787960: RNA sequencing of 6 HGSOC and 5 FTE. Open source

[3] Dou Z, Qiu C, Zhang X, et al. HJURP Promotes Malignant Progression and Mediates Sensitivity to Cisplatin and WEE1-inhibitor in Serous Ovarian Cancer. International Journal of Biological Sciences. 2022;18(3):1188-1210. doi:10.7150/ijbs.65589. Open source

[4] NCBI SRA. SRX13379937 / GSM5729031 / SRR17199566 (HGSOC1). Open source

[5] NCBI SRA. SRX13379943 / GSM5729037 / SRR17199572 (FT1). Open source

[6] NCBI SRA. SRX13379945 / GSM5729039 / SRR17199574 (FT3). Open source

[7] Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biology. 2014;15:550. doi:10.1186/s13059-014-0550-8. Open source

[8] Liao Y, Smyth GK, Shi W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. Bioinformatics. 2014;30(7):923-930. doi:10.1093/bioinformatics/btt656. Open source

Project-specific settings and file paths were drawn from the saved GSE190688 team settings checklist, DESeq2 preflight checklist, DESeq2 script and project summaries. Unconfirmed settings are marked for verification rather than assumed.